

First, reproduce learning in a simpler setting.

A recursive language model can use code and ask helper models to solve parts of a problem. We eventually want to train those choices.

Today's question

Can we use a published training recipe to improve a model's math answers on questions excluded from training?

We use the authors' code for **Dr. GRPO**, which learns from scores assigned to the model's own answers.

This is a small reproduction of an existing method. It establishes a reference for future work on recursive models.

Liu et al., *Understanding R1-Zero-Like Training: A Critical Perspective* (2025).

Official code; our run is much smaller than the paper's experiments.

Reward correct attempts, then change the model.

Illustrative question: Solve $3x + 2 = 14$. The correct answer is $x = 4$.

The model's attempts	Checker score	Learning direction
Two answers end in $x = 4$.	1 each	Make these more likely.
Two end in $x = 5$ or $x = 6$.	0 each	Make these less likely.

Each attempt is compared with the average score for that question. Training changes the model's weights using these differences.

The checker judges the final answer. We do not supply worked solutions for the model to copy.

Simplified illustration with four attempts; our runs use eight. If every attempt receives the same score, this question supplies no relative learning signal.

We train on math, then test on different questions.

We update the full weights of **Qwen2.5-Math-1.5B**, a pretrained math model, on one 40 GB A100 GPU.

Training run	Questions	Weight updates
Short pilot, completed	64	4
Larger run, restarted from the base model	512	32 planned

Training questions come from the authors' 12,000-question math collection. Test questions come from **MATH500**.

We use 64 questions to monitor progress and **128 different questions for the final test**. Neither group is used for training. Evaluation settings stay fixed before and after.

Evaluation chooses the most likely next token, up to 3,000 tokens. Exact question-overlap checks passed; pretraining exposure is unknown.

What changed on questions excluded from training?

Pilot: correct answers on 64 monitoring questions, after four updates.

Before training

20 / 64

After training

22 / 64

3 answers became correct; 1 became incorrect. An earlier evaluation of the same trained weights scored 23/64.

The larger comparison is still pending.

We will compare the base model with the final model after 32 updates on the separate 128-question test.

Authors' final checker throughout. This small, variable gain needs replication.

A higher score does not always mean a new skill.

An actual pilot example, shortened: How much fuel is saved over 12,000 miles by switching from 15 to 48 miles per gallon?

Before: repetition without an answer. After: a completed calculation.

$$12,000/15 - 12,000/48 = 800 - 250 = 550 \text{ gallons.}$$

Two of the pilot's three gains were recoveries from unfinished answers. One previously correct answer stopped finishing, too.

The lesson for RLMs: Check *what* training changes. Does the model merely finish reliably, or does it learn to ask useful subquestions?

The immediate next step is to repeat any promising result. This small experiment does not yet establish a reliable gain or reproduce the paper's full results.